# JIE GAO

高洁 (SHE/HER/HERS)

(My first name, "Jie", pronounced like "Jyeah" in English, and it means "clean and pure" in Chinese)

Information Systems Technology and Design Pillar
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## RESEARCH INTERESTS

PhD candidate with a background in Human-AI Interaction (HAI) and Computer-Supported Cooperative Work (CSCW). Passionate about leveraging AI and LLMs to enhance data analysis and create trustworthy and reliable AI-powered applications. Experienced in both quantitative and qualitative analysis for extracting data-driven insights, with a keen interest in applying these skills to cutting-edge and impactful research.

## EDUCATION

2019.09-present*
(*Expected in Jan 2024)

**Singapore University of Technology and Design**, *Singapore*
Ph.D. Student in Computer Science
Advisors: Simon Perrault, Kenny Choo
Committee: Wei Lu, Kwan Hui Lim, Darren Lee

2017.09-2019.06

**Zhejiang University**, *Hangzhou, China*
M.Eng. in Software Engineering
Advisors: Cheng Yao and Leijing Zhou

2013.09-2017.06

**Shanghai University**, *Shanghai, China*
B.Eng. in Digital Media Technology
Advisors: Feng Tian and Zhifeng Xie

## RESEARCH EXPERIENCES

2023.01-2023.07

**University of Notre Dame**, *IN, USA*
Visiting Ph.D. Student
Advisor: Toby Jia-jun Li

2022.06-2023.01

**National University of Singapore**, *Singapore*
Visiting Ph.D. Student
Advisor: Shengdong Zhao

## PUBLICATIONS (In Progress)

**Jie Gao**, Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, Simon Tangi Perrault. *CollabCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis*. arXiv preprint arXiv:2304.07366. Under review on *ACM Conference on Human Factors in Computing Systems (CHI24)*

**Jie Gao**, Junming Cao, Yeo Shunyi, Kenny Choo, Zheng Zhang, Toby Jia-Jun Li, Shengdong Zhao, Simon Tangi Perrault. *Impact of Qualitative Coding Strategies on User Trust, Reliance, and Helpfulness of AI-assisted Qualitative Coding System.* In submission.

ShunYi Yeo, Gionnieve Lim, **Jie Gao**, Weiyu Zhang, Simon Tangi Perrault. *Help Me Reflect: Leveraging Self-Reflection Interface Nudges to Enhance Deliberativeness on Online Deliberation Platforms.* Under review on *ACM Conference on Human Factors in Computing Systems (CHI24)*

ShunYi Yeo, Gionnieve Lim, **Jie Gao**, Simon Tangi Perrault. *Not Too Long, Not Too Short: Goldilocks Principle of 'Optimal' Time in Deliberation.* Major Revision from *ACM Conference On Computer-Supported Cooperative Work And Social Computing (CSCW2024)*

## PUBLICATIONS (Full Papers)

**Jie Gao**, Kenny Tsu Wei Choo, Junming Cao, Roy Ka Wei Lee, Simon Perrault. *CoAIcoder: Examining the Effectiveness of AI-assisted Human-to-Human Collaboration in Qualitative Analysis.* ACM Transactions on Computer-Human Interaction (TOCHI2023)

Zheng Zhang, **Jie Gao**, Ranjodh Singh Dhaliwal, and Toby Jia-Jun Li. *VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping.* ACM Symposium on User Interface Software and Technology (UIST2023)

Nuwan Janaka, **Jie Gao**, Lin Zhu, Shengdong Zhao, Lan Lyu, Peisen Xu, Maximilian Nabokow, Silang Wang, Yanch Ong. *GlassMessaging: Towards Ubiquitous Messaging Using OHMDs.* Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT2023)

Junming Cao, Bihuan Chen, Longjie Hu, **Jie Gao**, Kaifeng Huang, Xin Peng. *Understanding the Complexity and Its Impact on Testing in ML-Enabled Systems.* The International Conference on Software Maintenance and Evolution (ICSME2023)

## PUBLICATIONS (Short Paper/Workshop/Poster/Preprint)

**Jie Gao,** Yuchen Guo, Toby Jia-jun Li, Simon Perrault. *CollabCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis.* ACM Conference On Computer-Supported Cooperative Work And Social Computing *(CSCW Demo 2023)*

**Jie Gao,** Xiayin Ying, Junming Cao, Yifan Yang, PinSym Foong, Simon Perrault. *Differences of Challenges of Working from Home (WFH) between China and the US during COVID-19. ACM Conference on Human Factors in Computing Systems (CHI EA 2022)*

**Jie Gao,** Leijing Zhou, Miaomiao Dong, Fan Zhang. *Expressive Plant: A Multisensory Interactive System for Sensory Training of Children with Autism.* (Ubicomp Poster 2018)

**Jie Gao,** Pin Sym Foong, Yifan Yang, Weilin Jiang, Yijie Chen, Xiayin Ying, Simon Perrault. *From 996 to 007: Challenges of Working from Home During the Epidemic in China*. arXiv e-prints, arXiv: 2201.09045.

## ACADEMIC SERVICES

- Associate Chair for CHI Late-breaking Work 2023
- Associate Chair for CHI Late-breaking Work 2022
- Reviewers for CSCW2022, CSCW2023, CHI2023, AIED2023, CHI2024

## TEACHING ASSISTANT

- Introduction to Algorithms, Singapore University of Technology and Design, 2020 Fall
    - Undergraduate Course
    - Instructors: Kai Fong Ernest Chong, Ioannis Panageas
- Graphics & Visualization, Singapore University of Technology and Design, 2020 Summer
    - Undergraduate Course
    - Instructor: Peng Song

## MENTORING EXPERIENCE

- Shunyi Yeo (PhD Student, Singapore University of Technology and Design)
- Yifan Yang (PhD Student, University of Nottingham Ningbo China)
- Yuchen Guo (Undergraduate, Singapore University of Technology and Design)
- Tianqin Zhang (Undergraduate, Singapore University of Technology and Design)

## TECHNOLOGY SKILLS

- **Quantitative Data Analysis**: Numpy, Pandas, Matplotlib, Seaborn, SciPy, etc.
- **Qualitative Data Analysis:** Grounded Theory, Thematic Analysis, Atlas.ti
- **Frontend**: JavaScript, React.js, Nodejs, Html, CSS
- **Statistics and NLP**: SciPy, scikit-learn, PyTorch, NLTK, etc.
- **Design Skills:** Photoshop, Illustrator, Premiere
- **Others:** MongoDB, Git, Latex

[Updated as of December, 2023]